higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

INSTRUMENTATION TECHNOLOGY
NQF LEVEL 4

(6060474)

18 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 7 pages.

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TIME: 3 HOURS
MARKS: 150

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.5) in the ANSWER BOOK.

1.1.1 Which ONE of the following is NOT an element of SCADA?

A Remote telemetering unit
B Communication
C Human machine interface
D Distributed control system

1.1.2 Methods used for ultrasonic level measurement includes the … method.

A sonar
B frequency difference
C pressure difference
D hot-wire

1.1.3 A solution with a pH value of less than seven means that it is …

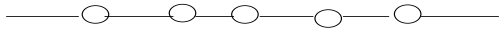
A neutral.
B alkaline.
C basic.
D acidic.

1.1.4 Piping and connections are represented with several different symbols.

Which symbol represents a data link?

A

B

C 

D 

1.1.5 A … transmitter can be used for level measurement of low viscosity liquids.

A standard DP
B flanged DP
C remote seal
D standard seal

(5 × 2) (10)

1.2 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–H) next to the question number (1.2.1–1.2.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.2.1	Chemical composition is determined by a desired chemical reaction and measures the electrolytic conductance	A	reference electrode
1.2.2	The resetting of the set point of a secondary controller by the output of a primary controller	B	feed-forward control
1.2.3	It provides a stable potential against which the measuring electrode voltage can be compared	C	SCADA
1.2.4	It monitors, controls and alarms the plant or regional operating systems from a central location	D	deviation
1.2.5	A system comprising a high-speed communications network, operations work center and microprocessor-based controllers	E	purge interlock
		F	cascade control
		G	dissolved-oxygen analyser
		H	DCS

(5 × 1) (5)

1.3 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.3.1–1.3.5) in the ANSWER BOOK.

1.3.1 A measuring electrode is sensitive to the hydrogen ions.

1.3.2 No comparison is made between the actual value and the desired value of a process variable in an open-loop system.

1.3.3 The second letter inside the circle in instrument labelling indicates a measured variable.

1.3.4 The density of any substance is directly proportional to the volume of that substance and inversely proportional to the mass.

1.3.5 The acronym P&ID stands for piping and instrumentation drawings.

(5 × 1) (5)

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1.4 Give ONE term for each of the following descriptions. Write only the term next to the question number (1.4.1–1.4.5) in the ANSWER BOOK

1.4.1 It is the measure of an object's resistance to acceleration (a change in its state of motion) when a net force is applied

1.4.2 Signals produced by changing the air pressure in a single pipe in proportion to the measured change in a process

1.4.3 A method of level monitoring by measuring the pressure in a virtually static liquid at a predetermined level of submersion

1.4.4 A quantity representing the force exerted on a particle or object by an acceleration field, particularly the gravitational field of the earth at the surface

1.4.5 The cloudiness or haziness of a fluid caused by suspended solids that are generally invisible to the naked eye

(5 × 2) (10)
[30]

TOTAL SECTION A: 30

SECTION B

QUESTION 2

2.1 Make a neat, labelled sketch of a typical installation arrangement of a closed pressurised vessel with non-volatile fluid, where a DP transmitter is used. (6)

2.2 Explain why a standard DP transmitter is not recommended in high viscosity liquids (liquids with solids in suspension). (3)

2.3 Make a neat, labelled sketch of an ultrasonic level measuring system. (5)

2.4 Discuss the principle of operation of an ultrasonic level detector. (7)

2.5 Name and explain THREE system configurations for open-tank level measurement. (3 × 2) (6)

2.6 Explain the measuring principle upon which all microwave level measuring instruments are based. (6)
[33]

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QUESTION 3

3.1 Name FIVE types of viscometers. (5)

3.2 Paramagnetic oxygen analysers are used in industry to determine the oxygen content in the product.

3.2.1 Explain the *paramagnetic property of gases* and the *diamagnetic property of gases*. (5)

3.2.2 Name TWO applications of the above-mentioned analyser. (2)

3.3 The moisture content of the atmosphere is of vital importance in many industrial processes.

3.3.1 Make a neat, labelled sketch of the electrical dew cell analyser. (8)

3.3.2 Explain the principle of operation of the analyser in QUESTION 3.3.1. (10)
[30]

QUESTION 4

4.1 Load cells are commonly used in the determination of the level of solid substances, such as the weighing of the material in a storage bin or tank.

4.1.1 Make a neat, labelled sketch showing how load cells are commissioned to measure the level of solid substances. (5)

4.1.2 Briefly discuss the principle of operation of load cells as used in QUESTION 4.1.1. (7)

4.2 Describe the maintenance requirements of load cells. (5)

4.3 State THREE criteria according to which load cells are classified and give an example of each. (3 × 2) (6)

4.4 State FOUR factors affecting the calibration of a mass meter employing the vibrating tube technique. (4)
[27]

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QUESTION 5

5.1 Describe the following configurations as used in control systems:

5.1.1 Fail safe open

5.1.1 Fail safe close

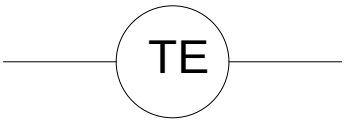
(2 × 2) (4)

5.2 State FIVE functions of statistical process control. (5)

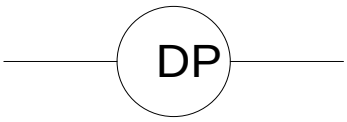
5.3 Name FIVE subsystems that a SCADA system usually consists of. (5)

5.4 What is the meaning of the following acronym labels?

5.4.1

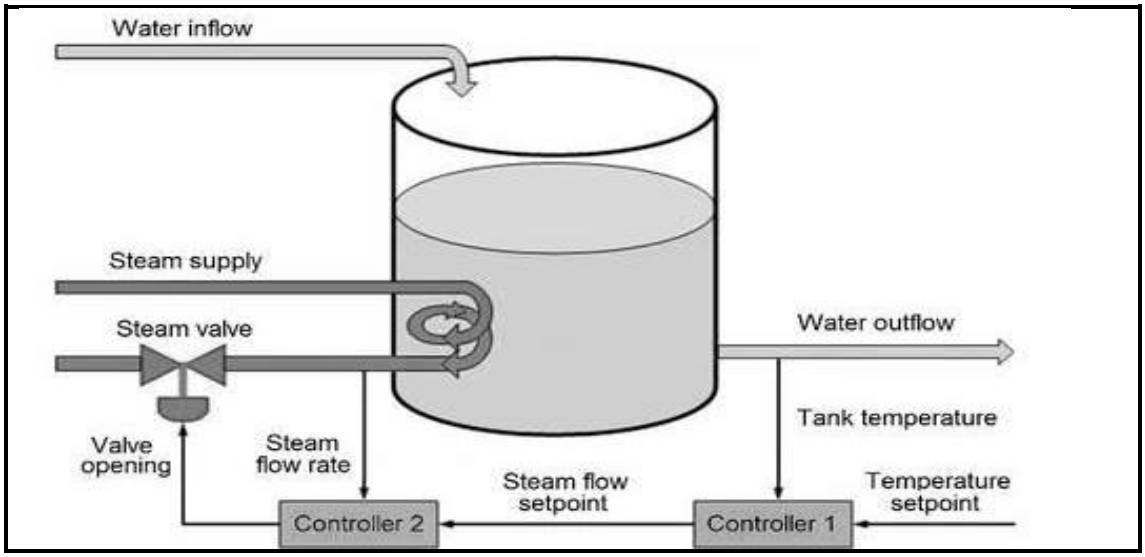


5.4.2



(2 × 3) (6)

5.5 Study the diagram of the control loop below and answer the questions.



5.5.1 What type of control loop is indicated by the diagram? (2)

5.5.2 Explain how the control loop in the diagram above regulates the water temperature. (8)

[30]

TOTAL SECTION B: 120
GRAND TOTAL: 150

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